The Journal of Open Source Software

rcrisp: An R package for urban river space delineation

Claudiu Forgaci [1*¶] **and Francesco Nattino** [2*]

1 Delft University of Technology, Julianalaan 134, 2628 BL Delft, The Netherlands **2** Netherlands eScience Center, Science Park 140, 1098 XG Amsterdam, The Netherlands ¶ Corresponding author * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 20 October 2026
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0)

Summary

`rcrisp` provides tools to automate the morphological delineation of riverside urban areas—i.e., the spaces surrounding a river where social, environmental and economic phenomena are assumed to be influenced by the presence of the river—following a method developed in Forgaci (2018, pp. 88–89). The method is based on the premise that analyses of riverside urban phenomena are often done without a clear and consistent spatial definition of the area of interest and that a morphological delineation—combining the natural terrain of the river valley and the configuration of the overlapping built urban fabric—can provide a more objective and comparable approach. Delineations following this morphological delineation method enable integrated local analyses, where data from different sources must be combined within a neutral spatial unit, as well as global cross-case analyses, which require a spatial unit that is comparable across different geographic contexts. The method proposes a hierarchical delineation of four spatial units: the river valley, the river corridor, the corridor segments, and the river space (i.e., the area between the riverbanks and the first line of buildings). Accordingly, `rcrisp` includes functions to delineate each, as well as an all-in-one function that runs all desired delineations. The package also includes functions to download and preprocess OpenStreetMap (OSM) and global Digital Elevation Model (DEM) data, which are required as input data for the delineation process. The resulting delineations can be used in any downstream analysis of riverside urban areas that can benefit from consistent and comparable spatial units, including land use, accessibility, and ecosystem services assessments.

Statement of need

Urban rivers and the urban areas surrounding them, hereinafter referred to as urban river spaces (URS), have come to the forefront of sustainable urban transformations worldwide. This trend is not only due to pounding water-related disturbances such as floods or water scarcity, but also because the vital role of rivers in supporting social and ecological processes is increasingly acknowledged. As researchers and practitioners of the urban environment engage into understanding or tackling current URS transformations, they face the challenge of capturing the complexities of environmental, social, and ecological phenomena in an integrated way (Prominski et al., 2023). Moreover, considering the global extent of the phenomenon, they lack established frames of comparison that may reveal local specificities and cross-case similarities. An essential part of the challenge is how urban areas surrounding rivers are delineated—that is, how their boundaries are drawn for analytical or intervention purposes—, as the resulting spatial units can considerably impact decisions made for urban riverspace transformations. More often than not, URS boundaries are determined without a systematic and reproducible delineation method, wherever suitable spatial data is available, based on census tracts, using rivers as boundaries, using neutral (square or circular) cut-outs, at different levels of aggregation (city, district, neighbourhood), or areas of study specific to a local

DRAFT

urban development project. None of these approaches to delineation are suitable for a reliable integrated understanding of URSs, as they are to a large extent arbitrary. An integrated understanding of URS requires spatial morphological units of analysis that reflect the combined "footprints" of environmental, social and ecological processes (Marcus & Pont, 2016). This is a case of the modifiable areal unit problem (MAUP) (Openshaw, 1984) with a considerable impact on decisions taken in environmental, mobility, ecological, and public space planning and design. Accordingly, `rcrisp` is designed to address a diverse, growing and cross-disciplinary research community concerned with understanding URSs in an integrated and scalable way. It is intended to open new research avenues, such as integrated local spatial analyses and global cross-case analyses, which have been so far hindered by data- or workflow-related difficulties.

State of the field

Most current software which may be usable for URS delineation purposes either rely on GUI-based GIS software that involve several manual steps that hinder scalability and reproducibility, or they are too generic for specialized applications such as URS delineation and analysis. Urban morphometrics, a growing sub-field of urban morphology, focuses mostly on the analysis of street networks, such as the Python package *OSMnx* (Boeing, 2025) and the R package `sfnetworks` (Meer et al., 2024), or on elements of built form, such as the Python toolkit *momepy* (Fleischmann, 2019; Fleischmann et al., 2022). These tools are not fit for specialised applications in URSs as they exclusively focus on generic elements of urban form and as such do not integrate green spaces and river spaces. On the other hand, river space delineation methods and software, like the one developed by the *Vermont Agency of Natural Resources* (2007) and the ATRIC tool developed by Bhowmik et al. (2015), mostly focus on non-urban contexts on a larger scale, where reading geomorphological information from digital elevation models in the absence of buildings and urban infrastructure is easier. Also, such specialised tools are either not open or they are not actively maintained.

Software design

`rcrisp` is an R package that builds on a previously developed delineation method Forgaci (2018, pp. 88–89), which considers the morphology of the river valley and that of the urban fabric jointly. `rcrisp` automates that method of delineation, while leveraging global geospatial data—OpenStreetMap (*OpenStreetMap*, 2025) and Global DEM (Copernicus, 2019) data—increasingly available for large-scale analyses (Forgaci, 2020).

`rcrisp` is structured into four core delineation modules (Figure 1), namely `valley`, `corridor`, `segments`, and `riverspace` for the four delineation steps, as well as an all-encompassing `delineate` module which runs the entire delineation workflow. The output is an object of class `delineation`. The delineation process is implemented using the following algorithms:

1. The `valley` module implements a cost distance algorithm to determine the river valley boundaries from a DEM, variants of which are mostly used for wet area mapping and valley bottom delineation in non-urban contexts (Ågren et al., 2014; Murphy et al., 2009; White et al., 2012).
2. The `corridor` module uses a shortest path algorithm to determine the corridor edges on the street network obtained from OpenStreetMap. For both sides of the river, the shortest paths which are closest to the valley boundaries are chosen for the corridor edge.
3. The `segments` module uses the COINS algorithm (Tripathy et al., 2020) implemented in the rcoins R package (Nattino et al., 2026) to determine continuous lines along the street network starting from bridges crossing the river. The resulting continuous lines are used to divide the corridor into corridor segments.

DRAFT

4. The `riverspace` module applies an isovist algorithm implemented in the visor R package (Forgaci & Nattino, 2026) to determine the space enclosed by the first line of buildings around the river within a given visibility distance.

An `aoi` module is used to set location parameters as input for delineation. The dedicated `network` and `sf` modules contain functionality needed for processing `sfnetwork` (Meer et al., 2024) and `sf` (Pebesma, 2018) objects, respectively. The retrieval of OSM data is available from the `osm` module, while example delineation data and example OSM data used in the documentation are available from the `data` and `exampledata` modules. Utility functions are collected in a `utils` module and the `cache` module is used to retain and manage OSM and DEM data retrieved by the user.

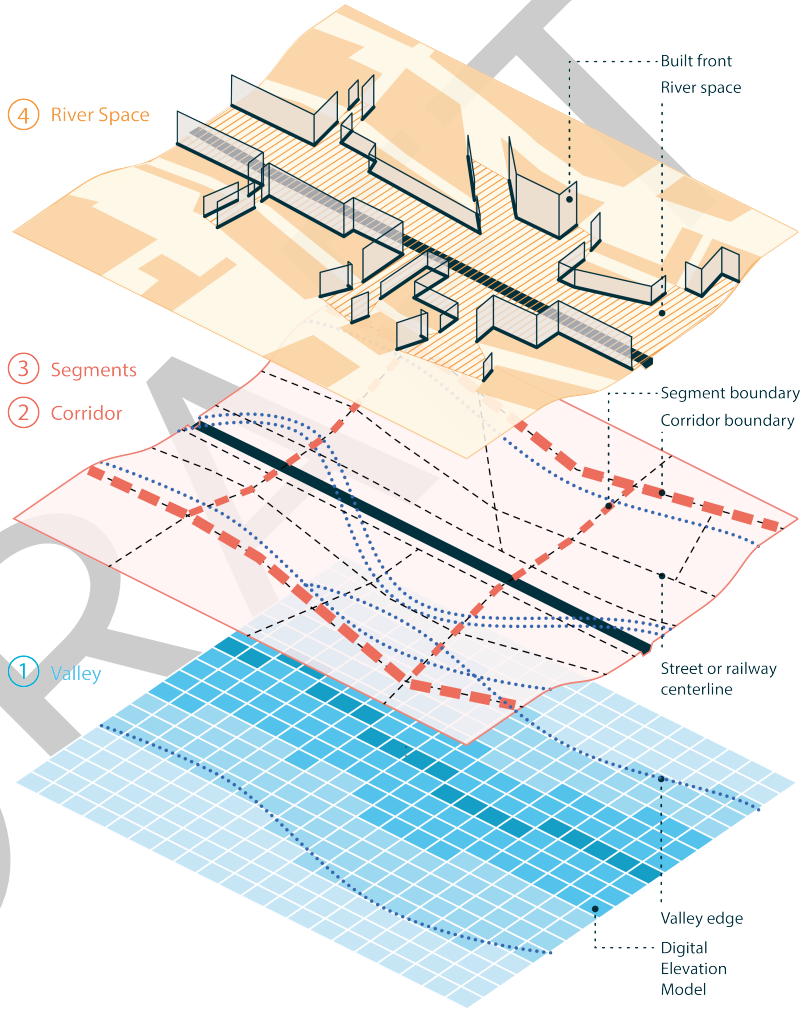


Figure 1: Diagram of the four delineation steps carried out by rcrisp.

A typical workflow consists of the following steps:

1. Define area of interest and parameters for a given city and a given river crossing it, using their names as input.
2. Get OSM and DEM base layers within the area of interest.
3. Run the all-in-one `delineate()` or delineation-specific `delineate_*()` functions to compute valley, corridor, segments, and/or riverspace. As an alternative to points 1-3,

run the convenience function `delineate_city_river()`.

4. Inspect and visualize the resulting delineation using `summary()` and `plot()` methods specific to the `delineation` class.
5. Export results in a desired geospatial format for downstream analysis.

Research impact statement

The significance of `rcrisp` has been demonstrated during early development as part of an interdisciplinary Lorentz Workshop in Leiden, the Netherlands, which brought together an international group of researchers, practitioners and students concerned with sustainable urban riverspace transformations. During the workshop, delineations for 17 international cases were validated and various use cases were discussed. Broad applications beyond rivers, i.e., for waterways in general (including canals and streams), as well as non-waterway-related use cases (e.g., urban catchment area along a given route) were also identified. A position paper (Forgaci et al., 2026) resulting from the workshop presents `rcrisp` as a catalyst for the development of a community around morphological urban riverspace transformations. The use of `rcrisp` has also been confirmed in research (Wu et al., 2026) and education (Hao, 2026) and the software continues to be employed in both.

AI usage disclosure

GitHub Copilot was used to expedite the review of GitHub pull requests. Claude Sonnet 4.6, Claude Opus 4.6, and Claude Opus 4.7 were used for test scaffolding in order to increase test coverage and identify edge cases, dummy data creation for tests, debugging, and the initial drafting of a benchmarking script. No other generative AI tools were used in the development of this software or the writing of this manuscript. The AI-generated code has been verified for correctness, accuracy and completeness, adapted where needed, and approved by the authors.

Acknowledgements

`rcrisp` was funded by the Netherlands eScience Center (grant no. NLESC.OEC.2023.050). We acknowledge contributions from Fakhereh Alidoost, Meiert Willem Grootes, and Yehan Wu.

References

Ågren, A. M., Lidberg, W., Strömgren, M., Ogilvie, J., & Arp, P. A. (2014). Evaluating digital terrain indices for soil wetness mapping – a swedish case study. *Hydrology and Earth System Sciences*, *18*(9), 3623–3634. https://doi.org/10.5194/hess-18-3623-2014

Bhowmik, A. K., Metz, M., & Schäfer, R. B. (2015). An automated, objective and open source tool for stream threshold selection and upstream riparian corridor delineation. *Environmental Modelling & Software*, *63*, 240–250. https://doi.org/10.1016/j.envsoft.2014.10.017

Boeing, G. (2025). Modeling and Analyzing Urban Networks and Amenities With OSMnx. *Geographical Analysis*, *57*(4), 567–577. https://doi.org/10.1111/gean.70009

Copernicus. (2019). *Copernicus DEM - Global and European Digital Elevation Model | Copernicus Data Space Ecosystem*. https://doi.org/10.5270/ESA-c5d3d65

Fleischmann, M. (2019). momepy: Urban Morphology Measuring Toolkit. *Journal of Open Source Software*, *4*(43), 1807. https://doi.org/10.21105/joss.01807

DRAFT

Fleischmann, M., Feliciotti, A., & Kerr, W. (2022). Evolution of Urban Patterns: Urban Morphology as an Open Reproducible Data Science. *Geographical Analysis*, *54*(3), 536–558. https://doi.org/10.1111/gean.12302

Forgaci, C. (2018). *Integrated urban river corridors: Spatial design for social-ecological integration in bucharest and beyond* [PhD thesis]. https://doi.org/10.7480/abe.2018.31

Forgaci, C. (2020). Smart and resilient cities: How can big data inform spatial design and planning for urban resilience? *Contesti. Città, Territori, Progetti*, *1*, 62–71. https://doi.org/10.13128/contest-12035

Forgaci, C., & Nattino, F. (2026). *Visor: Geospatial tools for visibility analysis*. https://doi.org/10.5281/zenodo.15133420

Forgaci, C., Nattino, F., Stan, A., Alidoost, F., Montezuma, M. A., Bektaș, B. N., de Boer, L., Cannatella, D., d'Avila, R., Ellenbroek, I., Goiati, I., Haase, U., Hausleitner, B., Joh, H., van der Kwast, J., Mišík, M., Mukkamala, T., Negrello, M., Patchineelam, S. M., … Wu, Y. (2026). *Sustainable Urban River Space Transformations: An Interdisciplinary Spatial Agenda for Research and Practice* (No. crw3h_v1). SocArXiv. https://doi.org/10.31235/osf.io/crw3h_v1

Hao, Y. (2026). *Towards a Resilient Port-Cityscape: Identifying waterfront transformation potential in industrial inland port cities in transition in the hinterland of Port of Rotterdam* [PhD thesis]. https://repository.tudelft.nl/record/uuid:3d6d3156-7855-41d9-97b6-da268a00c5cd

Marcus, L., & Pont, M. B. (2016). Towards a social-ecological urban morphology: integrating urban form and landscape ecology. *Proceedings: ISUF 2015 XXII International Conference: City as Organism: New Visions for Urban Life*, *1*(1), 371–378. https://research.chalmers.se/en/publication/247483

Meer, L. van der, Abad, L., Gilardi, A., & Lovelace, R. (2024). *Sfnetworks: Tidy geospatial networks*. https://doi.org/10.32614/CRAN.package.sfnetworks

Murphy, P. N. C., Ogilvie, J., & Arp, P. (2009). Topographic modelling of soil moisture conditions: a comparison and verification of two models. *European Journal of Soil Science*, *60*(1), 94–109. https://doi.org/10.1111/j.1365-2389.2008.01094.x

Nattino, F., Forgaci, C., & Alidoost, F. (2026). *Rcoins: Identify naturally continuous lines in a spatial network*. https://doi.org/10.5281/zenodo.14501804

Openshaw, S. (1984). *The modifiable areal unit problem*. Geo.

OpenStreetMap. (2025). https://www.openstreetmap.org/about

Pebesma, E. (2018). The R Journal: Simple Features for R: Standardized Support for Spatial Vector Data. *The R Journal*, *10*(1), 439–446. https://doi.org/10.32614/RJ-2018-009

Prominski, M., Stokman, A., Stimberg, D., Voermanek, H., Zeller, S., Bajc, K., & Zheng, N. (2023). *River.Space.Design: Planning Strategies, Methods and Projects for Urban Rivers Third and Enlarged Edition*. Birkhäuser. https://doi.org/10.1515/9783035625271

Tripathy, P., Rao, P., Balakrishnan, K., & Malladi, T. (2020). An open-source tool to extract natural continuity and hierarchy of urban street networks. *Environment and Planning B: Urban Analytics and City Science*, *48*(8), 2188–2205. https://doi.org/10.1177/2399808320967680

Vermont Agency of Natural Resources. (2007). *Stream and River Corridor Delineation Process*. https://dec.vermont.gov/sites/dec/files/wsm/rivers/docs/assessment-protocol-appendices/E-Appendix-E-07-Corridor-Definition.pdf

White, B., Ogilvie, Campbell, Hiltz, Gauthier, Chisholm, Wen, Murphy, & Arp, P. A. A. and. (2012). Using the cartographic depth-to-water index to locate small streams and associated

DRAFT

wet areas across landscapes. *Canadian Water Resources Journal / Revue Canadienne Des Ressources Hydriques*, *37*(4), 333–347. https://doi.org/10.4296/cwrj2011-909

Wu, Y., Forgaci, C., Franzke, N., Haase, U., John, H., Mrozik, K. D., & Syrbe, R.-U. (2026). *Typology-based optimization reveals urban stream restoration priorities for biodiversity, climate adaptation and quality of life* (No. qb2x6_v1). SocArXiv. https://doi.org/10.31235/osf.io/qb2x6_v1

DRAFT